

NMEC595 - Research Methodology

Lecture 6: Measurement and Scaling Techniques

Prof. Suparno Bhattacharyya

suparno@iitism.ac.in

224A, Mechanical Engg. Building

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Dept. of Mechanical Engineering, IIT(ISM) Dhanbad

Lecture Outline

- 1 Understanding Measurement
- 2 Types of Data and Scales
- 3 Measurement Errors
- 4 Tests of Sound Measurement
- 5 Developing Measurement Tools
- 6 Scaling Techniques
- 7 Key Takeaways

What is Measurement?

Definition: Measurement means giving numbers to things or observations, following certain rules.

Important Points:

- We can measure **physical things** like length, weight, and temperature, and also **abstract ideas** like machine efficiency or vibration severity.
- Measuring is easy when we have standard units (like meters or kilograms), but harder for abstract concepts (like how smooth a machine runs).
- We are less sure about measurements when the idea is abstract and the tools are not clear or standard.

How it works: Measurement matches things we want to study (domain) with numbers (range) using a clear rule.

Examples:

- Length of a shaft: Object $\rightarrow$ Number (in millimeters)
- Temperature of a motor: Object $\rightarrow$ Number (in degrees Celsius)
- Vibration level: Machine condition $\rightarrow$ Number (scale from 1 = low vibration to 5 = high vibration)

Classification of Data

- Classify data as
 - Qualitative (categorical) or quantitative (numeric);
 - Quantitative (numeric): discrete, continuous, or neither; and
 - (Classification based on information content): nominal, ordinal, interval, or ratio.

Qualitative vs. Quantitative Data

Qualitative data, also known as *categorical* data, consist of labels or descriptions of traits.

Quantitative data, also known as numeric data, consist of counts/measurements.

Continuous vs. Discrete Data

Discrete data are quantitative data that can take on only particular values and are usually counts.

Continuous data are quantitative data that can take on any value in a given interval.

Levels of Measurement

1. Nominal Data (qualitative): Numbers are just labels

- You cannot do math with these numbers (e.g., you can't say 4 is greater than 2 or subtract them meaningfully)
- Example: Marital status coded as 1, 2, 3, 4; jersey numbers of basketball players

2. Ordinal Data (qualitative): Numbers show order or rank only

- You can say which is bigger or smaller, but you cannot measure the exact difference between them
- Example: Mohs hardness scale (1 to 10); you can say 5 is harder than 2, but the difference between 10 and 9 is not the same as between 5 and 4

3. Interval Data (quantitative): Numbers show order and exact differences

- Differences between values are meaningful and equal, but there is no true zero point
- Example: Temperature in Fahrenheit; the difference between 95°F and 70°F is the same as between 135°F and 110°F, but 126°F is not twice as hot as 63°F

4. Ratio Data (quantitative): Numbers have order, equal differences, and a true zero

- All math operations are valid, including ratios
- Example: Weight or length; 126 kg is exactly twice as heavy as 63 kg

Comparison of Measurement Scales

Scale	Key Features	Examples	Central Tendency	Common Tests
Nominal	Categories only, no order	Gender, Player numbers	Mode	Chi-square
Ordinal	Ranked order, unequal intervals	Class rank, Mohs hardness	Median	Non-parametric tests
Interval	Equal intervals, no true zero	Fahrenheit temperature, IQ scores	Mean, Standard Deviation	t-test, F-test
Ratio	True zero, meaningful ratios	Weight, Height	Mean, Geometric Mean	All statistical tests

Key Idea: As you move from nominal (least precise) to ratio (most precise), the amount of information and the types of analysis you can do increase.

Tip: Always use the most precise scale that fits your data. For example, physical sciences often use ratio scales, while behavioral sciences commonly use interval scales.

Four Sources of Measurement Error

Goal: Identify and manage errors to ensure accurate and reliable results.

1. Respondent Errors:

- Hesitation to share true feelings; may pretend to know when they don't
- Temporary conditions like tiredness, boredom, anxiety, or mood swings

2. Situation Errors:

- Environmental distractions such as noise, temperature, or poor timing
- Presence of others affecting honesty; worries about privacy; poor interviewer-respondent connection

3. Measurer Errors:

- Changing or reordering questions; interviewer's tone or behavior influencing answers
- Mistakes in data handling: wrong coding, errors in tabulation or calculations

4. Instrument Errors:

- Confusing or complex wording; unclear instructions; poor layout or printing
- Incomplete coverage of the topic; technical problems with measurement tools

Three Key Tests of Measurement Quality: Validity, Reliability, Practicality

Definition: Validity is the extent to which a measurement tool accurately measures what it is intended to measure.

Example: A weighing scale should measure *weight*. If it displays random numbers, it is not valid.

Context: Validity is the most critical test in measurement. Even if a tool is reliable (consistent results), it is not useful unless it is also valid (measures the correct concept).

Content Validity

Content Validity: Assesses whether the measurement covers all important aspects of the concept.

- **What it checks:** Does the tool cover every relevant area of the topic?
- **Example:** A biology exam should include questions on cells, genetics, ecology, and evolution, not only one topic.
- **How judged:** Experts or panels evaluate if the content is comprehensive.

Criterion-Related and Construct Validity

Criterion-Related Validity: Determines if a measurement tool predicts or reflects real-world outcomes.

- **Subtypes:**

- Predictive: Predicts future outcomes (e.g., entrance exam scores predicting college performance).
- Concurrent: Correlates with current, established measures (e.g., new and established depression tests administered together).

- **How measured:** Correlation coefficient

- **Criterion:** Must be relevant, unbiased, reliable, and accessible.

Construct Validity: Assesses whether a tool accurately measures an abstract concept, consistent with theory.

- **Example:** A test for "critical thinking" should truly reflect critical thinking skills.
- **Note:** Abstract constructs (e.g., intelligence) are not directly observable, making this type complex.

Reliability and Practicality

2. **Reliability**: Consistency of results

- A reliable instrument gives consistent results every time.
- Reliability supports validity, but a valid instrument is always reliable.
- Example: A scale that always adds 5 kg is reliable but not valid.
- Reliability is generally easier to check than validity.

Two Key Aspects of Reliability:

- **Stability**: Same results when measuring the same person or item repeatedly.
- **Equivalence**: Consistency between different testers or samples.

How to Improve Reliability: (i) Standardize testing conditions and reduce fatigue or boredom. (ii) Use trained personnel, clear instructions, and a wide range of test items.

3. **Practicality**: Ease of use and cost-effectiveness

- Balance between ideal measurement and available resources.
- Should be easy to administer with clear instructions and scoring.

Four-Stage Development Process

Stage 1: Concept Development

- Understand the main ideas related to your study
- Especially important in theoretical research

Stage 2: Specify Dimensions

- Identify key parts of the concept, either by intuition or data analysis
- Example (Company Image): Product reputation, customer service, leadership, care for employees, social responsibility

Stage 3: Select Indicators

- Choose specific questions or scales that measure knowledge, opinions, or expectations
- Use several indicators to improve accuracy and reliability
- Each indicator only partially reflects the true value

Stage 4: Form Index

- Combine all indicators into one overall score or index
- Use summed or averaged values
- This gives a better measurement than any single indicator alone

Introduction to Scaling

Definition: *“Methods of assigning numbers to different levels of opinions, attitudes, or other concepts.”*

Two Approaches:

- Directly judge the characteristic and place it on a scale
- Design questionnaires where the answers give a score that shows the position on the scale

What is a Scale? A scale is a range from the highest to the lowest point, with steps in between. Higher numbers mean a greater amount of the characteristic being measured.

Six Bases for Scale Classification

1. Subject Orientation:

- **Respondent-oriented:** Measures characteristics of the respondent
- **Stimulus-oriented:** Focuses on judging a specific object or stimulus

2. Response Form:

- **Categorical (Rating):** Scores given without comparing to others
- **Comparative (Ranking):** Objects are compared or ranked against each other

3. Degree of Subjectivity: Measures subjective preferences or objective, non-preference judgments

4. Scale Properties: Nominal, ordinal, interval, or ratio scales

5. Number of Dimensions: Unidimensional (single trait) vs. multidimensional (multiple traits)

6. Construction Techniques: (To be covered on the next slide)

Five Scale Construction Techniques

1. Arbitrary Approach:

- Items chosen in an ad hoc way, without strong evidence
- Most commonly used but lacks solid measurement support

2. Consensus Approach:

- A panel of experts reviews items for relevance and clarity

3. Item Analysis Approach:

- Give the test, calculate total scores
- Identify which items best separate high scorers from low scorers

4. Cumulative Scales:

- Items ranked by how well they discriminate
- Agreeing with an extreme statement means agreeing with all less extreme ones

5. Factor Scales:

- Uses factor analysis to group items based on how they correlate
- The most advanced and precise method

Summary: Core Principles

- **Measurement:** Assigning numbers to observations based on rules; mapping from domain to range.
- **Four Data/Scale Types:** Nominal (labels), Ordinal (rank), Interval (equal differences), Ratio (true zero).
- **Parametric vs Non-Parametric:** Parametric uses interchangeable units; non-parametric does not—this affects choice of statistical methods.
- **Four Sources of Error:** Respondent (fatigue, reluctance), Situation (environmental factors), Measurer (bias, data handling), Instrument (design flaws).
- **Three Key Tests:** Validity (most important: content, criterion-related, construct), Reliability (stability, equivalence), Practicality (cost, ease, clarity).
- **Development Process:** Four stages—concept development, specifying dimensions, selecting indicators, forming an index.
- **Scaling:** Assigning numbers to levels of opinion or attitude; classified by six bases.
- **Scale Construction:** Five techniques—arbitrary, consensus, item analysis, cumulative, and factor analysis.

Practical Guidance

For Researchers:

- 1 Choose the most precise scale that fits the nature of your variable.
- 2 Physical sciences typically use ratio scales; behavioral sciences often use interval scales.
- 3 Ensure your measurements are accurate and clearly defined.
- 4 Address all four sources of measurement error carefully.
- 5 Confirm your tools meet validity, reliability, and practicality standards.
- 6 Use multiple indicators to improve stability and validity.
- 7 Balance ideal research goals with budget and resource limits.
- 8 Validate your measurement tools before full-scale use.

Important Notes:

- Reliability does not guarantee validity, but validity always implies reliability.
- Increasing sample size cannot fix systematic measurement errors.